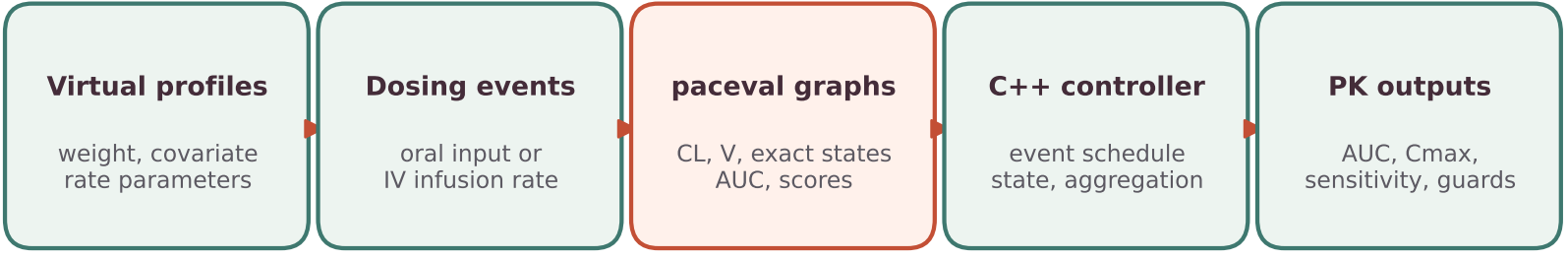Research workflow: persistent mathematical kernels in a synthetic PK simulation